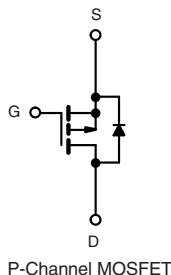
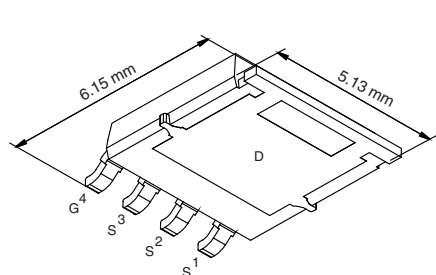


Automotive P-Channel 60 V (D-S) 175 °C MOSFET

PRODUCT SUMMARY

V_{DS} (V)	- 60
$R_{DS(on)}$ (Ω) at $V_{GS} = -10$ V	0.016
$R_{DS(on)}$ (Ω) at $V_{GS} = -4.5$ V	0.021
I_D (A)	- 30
Configuration	Single

PowerPAK® SO-8L Single



FEATURES

- Halogen-free According to IEC 61249-2-21 Definition
- TrenchFET® Power MOSFET
- AEC-Q101 Qualified^d
- 100 % R_g and UIS Tested
- Compliant to RoHS Directive 2002/95/EC



RoHS
COMPLIANT
HALOGEN
FREE

ORDERING INFORMATION

Package	PowerPAK SO-8L
Lead (Pb)-free and Halogen-free	SQJ461EP-T1-GE3

ABSOLUTE MAXIMUM RATINGS ($T_C = 25$ °C, unless otherwise noted)

PARAMETER	SYMBOL	LIMIT	UNIT
Drain-Source Voltage	V_{DS}	- 60	V
Gate-Source Voltage	V_{GS}	± 20	
Continuous Drain Current ^a	I_D	$T_C = 25$ °C	A
		$T_C = 125$ °C	
Continuous Source Current (Diode Conduction) ^a	I_S	- 30	
Pulsed Drain Current ^b	I_{DM}	- 120	
Single Pulse Avalanche Current	I_{AS}	- 50	
Single Pulse Avalanche Energy	E_{AS}	125	mJ
Maximum Power Dissipation ^b	P_D	$T_C = 25$ °C	W
		$T_C = 125$ °C	
Operating Junction and Storage Temperature Range	T_J, T_{stg}	- 55 to + 175	°C
Soldering Recommendations (Peak Temperature) ^{e, f}		260	

THERMAL RESISTANCE RATINGS

PARAMETER	SYMBOL	LIMIT	UNIT
Junction-to-Ambient	R_{thJA}	65	°C/W
Junction-to-Case (Drain)	R_{thJC}	1.8	

Notes

- Package limited.
- Pulse test; pulse width ≤ 300 μ s, duty cycle ≤ 2 %.
- When mounted on 1" square PCB (FR-4 material).
- Parametric verification ongoing.
- See Solder Profile (www.vishay.com/doc?73257). The PowerPAK SO-8L. The end of the lead terminal is exposed copper (not plated) as a result of the singulation process in manufacturing. A solder fillet at the exposed copper tip cannot be guaranteed and is not required to ensure adequate bottom side solder interconnection.
- Rework conditions: manual soldering with a soldering iron is not recommended for leadless components.



SPECIFICATIONS (T _C = 25 °C, unless otherwise noted)							
PARAMETER	SYMBOL	TEST CONDITIONS		MIN.	TYP.	MAX.	UNIT
Static							
Drain-Source Breakdown Voltage	V _{DS}	V _{GS} = 0 , I _D = - 250 μA		- 60	-	-	V
Gate-Source Threshold Voltage	V _{GS(th)}	V _{DS} = V _{GS} , I _D = - 250 μA		- 1.5	- 2.0	- 2.5	
Gate-Source Leakage	I _{GSS}	V _{DS} = 0 V, V _{GS} = ± 20 V		-	-	± 100	nA
Zero Gate Voltage Drain Current	I _{DSS}	V _{GS} = 0 V	V _{DS} = - 60 V	-	-	- 1	μA
		V _{GS} = 0 V	V _{DS} = - 60 V, T _J = 125 °C	-	-	- 50	
		V _{GS} = 0 V	V _{DS} = - 60 V, T _J = 175 °C	-	-	- 150	
On-State Drain Current ^a	I _{D(on)}	V _{GS} = - 10 V	V _{DS} ≥ 5 V	- 30	-	-	A
Drain-Source On-State Resistance ^a	R _{DS(on)}	V _{GS} = - 10 V	I _D = - 14.4 A	-	0.013	0.016	Ω
		V _{GS} = - 4.5 V	I _D = - 12.6 A	-	0.017	0.021	
		V _{GS} = - 10 V	I _D = - 14.4 A, T _J = 125 °C	-	0.021	0.026	
		V _{GS} = - 10 V	I _D = - 14.4 A, T _J = 175 °C	-	0.026	0.032	
Forward Transconductance ^b	g _{fs}	V _{DS} = - 15 V, I _D = - 14.4 A		-	40	-	S
Dynamic ^b							
Input Capacitance	C _{iss}	V _{GS} = 0 V	V _{DS} = - 30 V, f = 1 MHz	-	3920	4710	pF
Output Capacitance	C _{OSS}			-	420	510	
Reverse Transfer Capacitance	C _{rss}			-	295	360	
Total Gate Charge ^c	Q _g	V _{GS} = - 10 V	V _{DS} = - 30 V, I _D = - 14.4 A	-	90	140	nC
Gate-Source Charge ^c	Q _{gs}			-	13	-	
Gate-Drain Charge ^c	Q _{gd}			-	22	-	
Gate Resistance	R _g	f = 1 MHz		1.4	2.3	3.2	Ω
Turn-On Delay Time ^c	t _{d(on)}	V _{DD} = - 30 V, R _L = 30 Ω I _D ≅ - 1 A, V _{GEN} = - 10 V, R _g = 6 Ω		-	16	20	ns
Rise Time ^c	t _r			-	10	13	
Turn-Off Delay Time ^c	t _{d(off)}			-	70	85	
Fall Time ^c	t _f			-	22	30	
Source-Drain Diode Ratings and Characteristics ^b							
Pulsed Current ^a	I _{SM}			-	-	- 120	A
Forward Voltage	V _{SD}	I _F = - 4.5 A, V _{GS} = 0		-	- 0.8	- 1.2	V

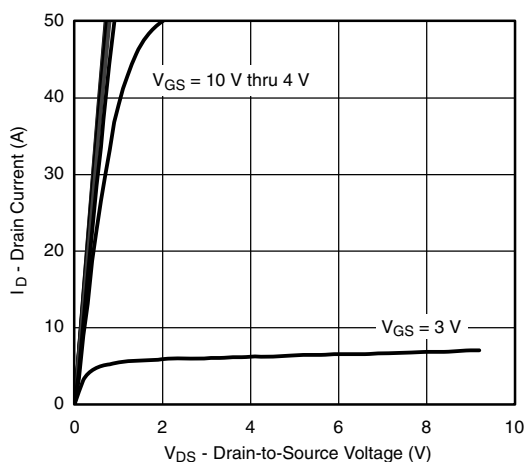
Notes

- a. Pulse test; pulse width $\leq 300\text{ }\mu\text{s}$, duty cycle $\leq 2\%$.
b. Guaranteed by design, not subject to production testing.
c. Independent of operating temperature.

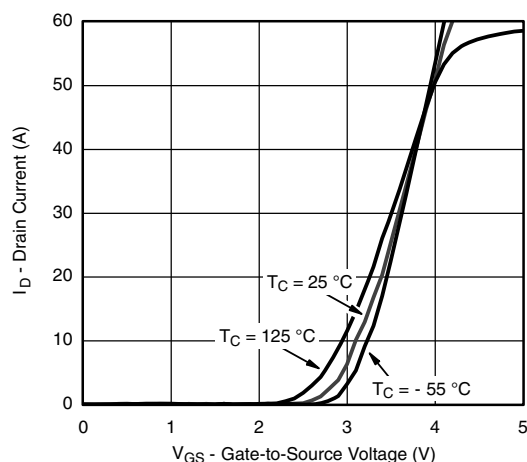
Stresses beyond those listed under "Absolute Maximum Ratings" may cause permanent damage to the device. These are stress ratings only, and functional operation of the device at these or any other conditions beyond those indicated in the operational sections of the specifications is not implied. Exposure to absolute maximum rating conditions for extended periods may affect device reliability.



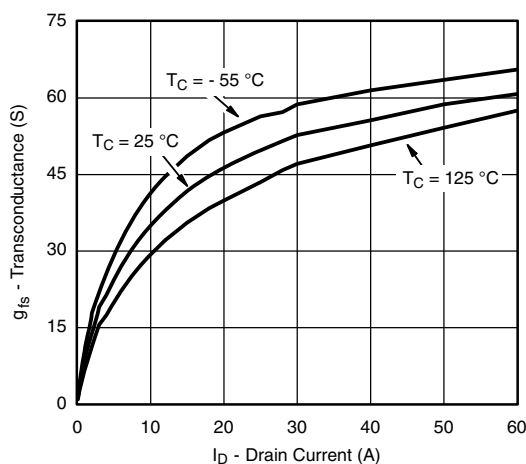
TYPICAL CHARACTERISTICS ($T_A = 25\text{ }^{\circ}\text{C}$, unless otherwise noted)



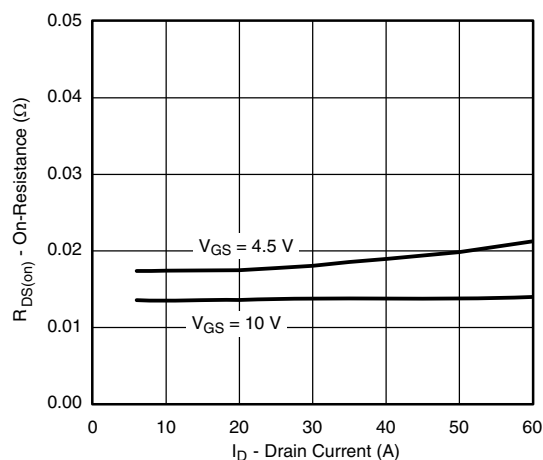
Output Characteristics



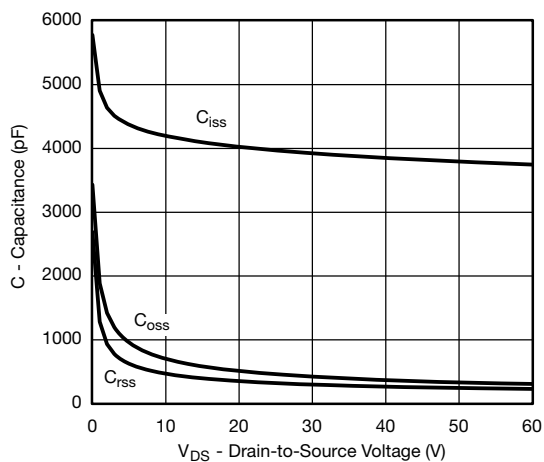
Transfer Characteristics



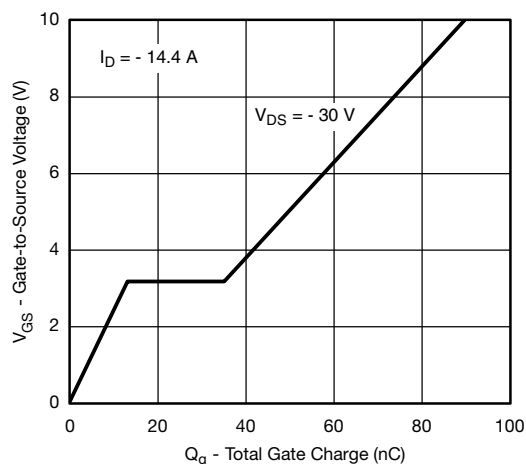
Transconductance



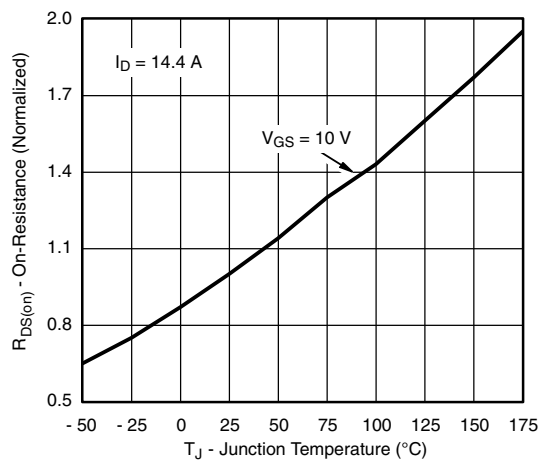
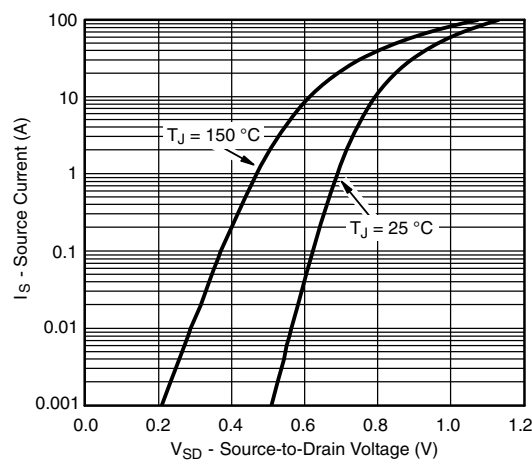
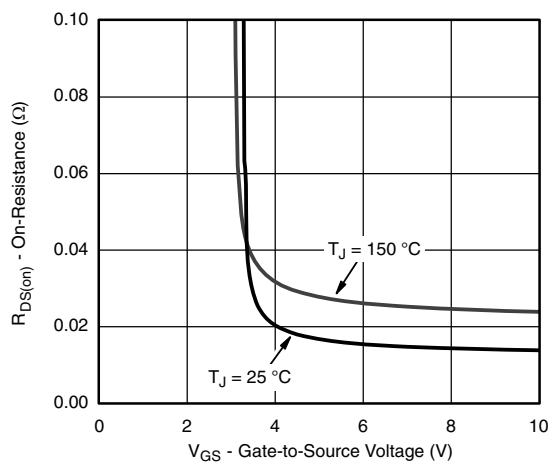
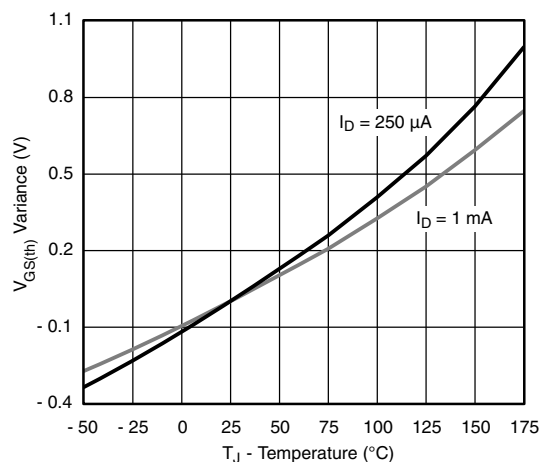
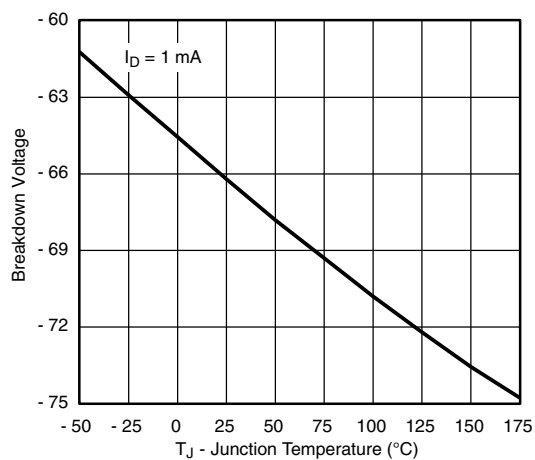
On-Resistance vs. Drain Current



Capacitance

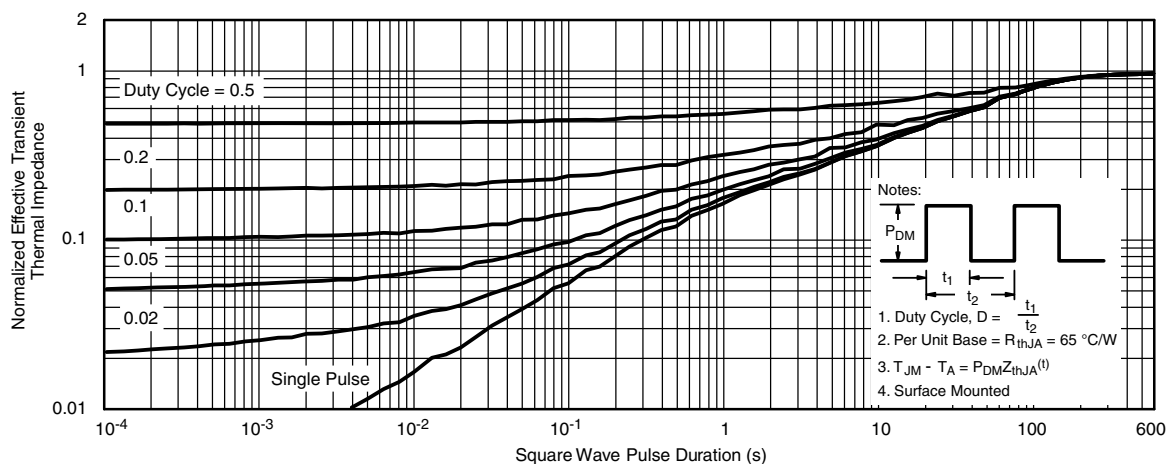
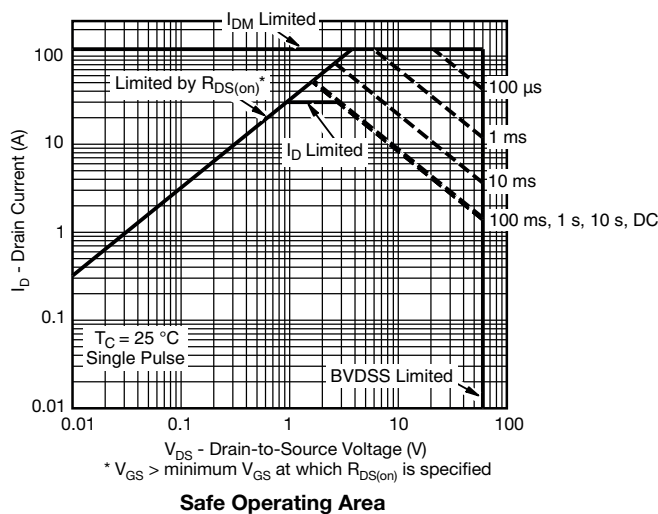


Gate Charge

TYPICAL CHARACTERISTICS ($T_A = 25\text{ }^{\circ}\text{C}$, unless otherwise noted)

On-Resistance vs. Junction Temperature

Source Drain Diode Forward Voltage

On-Resistance vs. Gate-to-Source Voltage

Threshold Voltage

Breakdown Voltage vs. Junction Temperature



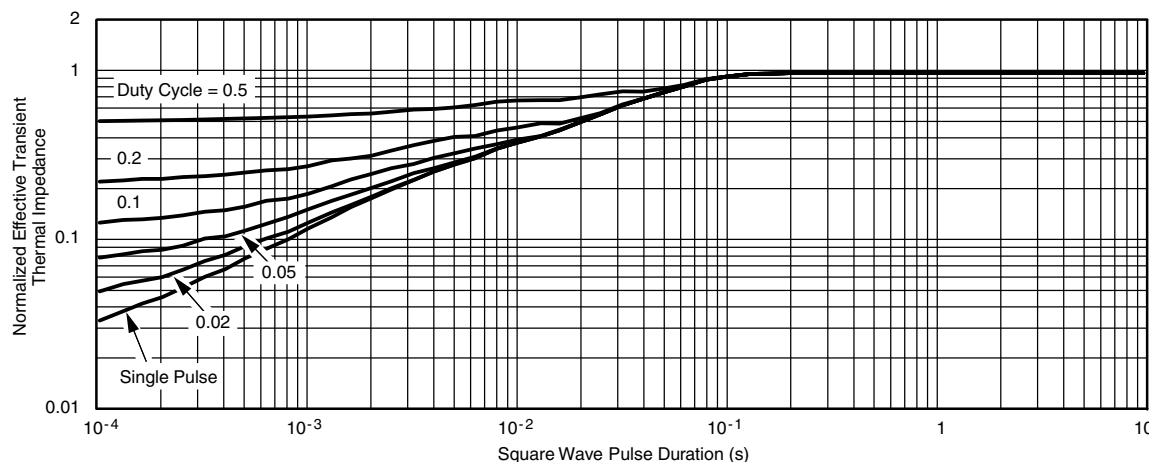
THERMAL RATINGS ($T_A = 25\text{ }^{\circ}\text{C}$, unless otherwise noted)



Normalized Thermal Transient Impedance, Junction-to-Ambient



THERMAL RATINGS ($T_A = 25\text{ }^{\circ}\text{C}$, unless otherwise noted)



Normalized Thermal Transient Impedance, Junction-to-Case

Note

- The characteristics shown in the two graphs
 - Normalized Transient Thermal Impedance Junction-to-Ambient ($25\text{ }^{\circ}\text{C}$)
 - Normalized Transient Thermal Impedance Junction-to-Case ($25\text{ }^{\circ}\text{C}$)are given for general guidelines only to enable the user to get a “ball park” indication of part capabilities. The data are extracted from single pulse transient thermal impedance characteristics which are developed from empirical measurements. The latter is valid for the part mounted on printed circuit board - FR4, size 1" x 1" x 0.062", double sided with 2 oz. copper, 100 % on both sides. The part capabilities can widely vary depending on actual application parameters and operating conditions.

Vishay Siliconix maintains worldwide manufacturing capability. Products may be manufactured at one of several qualified locations. Reliability data for Silicon Technology and Package Reliability represent a composite of all qualified locations. For related documents such as package/tape drawings, part marking, and reliability data, see www.vishay.com/ppg?65541.

**PowerPAK® SO-8L**

Ordering codes for the SQ rugged series power MOSFETs in the PowerPAK SO-8L package:

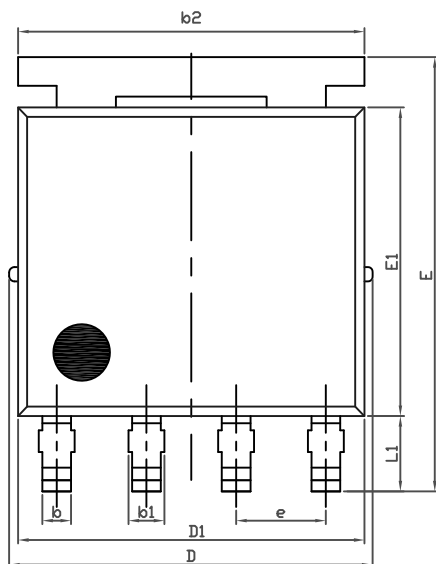
DATASHEET PART NUMBER	OLD ORDERING CODE ^a	NEW ORDERING CODE
SQJ401EP	SQJ401EP-T1-GE3	SQJ401EP-T1_GE3
SQJ402EP	SQJ402EP-T1-GE3	SQJ402EP-T1_GE3
SQJ403EEP	SQJ403EEP-T1-GE3	SQJ403EEP-T1_GE3
SQJ403EP	-	SQJ403EP-T1_GE3
SQJ410EP	SQJ410EP-T1-GE3	SQJ410EP-T1_GE3
SQJ412EP	SQJ412EP-T1-GE3	SQJ412EP-T1_GE3
SQJ422EP	SQJ422EP-T1-GE3	SQJ422EP-T1_GE3
SQJ431EP	SQJ431EP-T1-GE3	SQJ431EP-T1_GE3
SQJ443EP	SQJ443EP-T1-GE3	SQJ443EP-T1_GE3
SQJ456EP	SQJ456EP-T1-GE3	SQJ456EP-T1_GE3
SQJ460AEP	-	SQJ460AEP-T1_GE3
SQJ461EP	SQJ461EP-T1-GE3	SQJ461EP-T1_GE3
SQJ463EP	SQJ463EP-T1-GE3	SQJ463EP-T1_GE3
SQJ465EP	SQJ465EP-T1-GE3	SQJ465EP-T1_GE3
SQJ469EP	SQJ469EP-T1-GE3	SQJ469EP-T1_GE3
SQJ486EP	SQJ486EP-T1-GE3	SQJ486EP-T1_GE3
SQJ488EP	SQJ488EP-T1-GE3	SQJ488EP-T1_GE3
SQJ500AEP	SQJ500AEP-T1-GE3	SQJ500AEP-T1_GE3
SQJ840EP	SQJ840EP-T1-GE3	SQJ840EP-T1_GE3
SQJ844AEP	SQJ844AEP-T1-GE3	SQJ844AEP-T1_GE3
SQJ850EP	SQJ850EP-T1-GE3	SQJ850EP-T1_GE3
SQJ858AEP	SQJ858AEP-T1-GE3	SQJ858AEP-T1_GE3
SQJ886EP	SQJ886EP-T1-GE3	SQJ886EP-T1_GE3
SQJ910AEP	SQJ910AEP-T1-GE3	SQJ910AEP-T1_GE3
SQJ912AEP	SQJ912AEP-T1-GE3	SQJ912AEP-T1_GE3
SQJ940EP	SQJ940EP-T1-GE3	SQJ940EP-T1_GE3
SQJ942EP	SQJ942EP-T1-GE3	SQJ942EP-T1_GE3
SQJ951EP	SQJ951EP-T1-GE3	SQJ951EP-T1_GE3
SQJ952EP	-	SQJ952EP-T1_GE3
SQJ960EP	SQJ960EP-T1-GE3	SQJ960EP-T1_GE3
SQJ963EP	SQJ963EP-T1-GE3	SQJ963EP-T1_GE3
SQJ968EP	SQJ968EP-T1-GE3	SQJ968EP-T1_GE3
SQJ980AEP	SQJ980AEP-T1-GE3	SQJ980AEP-T1_GE3
SQJ992EP	SQJ992EP-T1-GE3	SQJ992EP-T1_GE3

Note

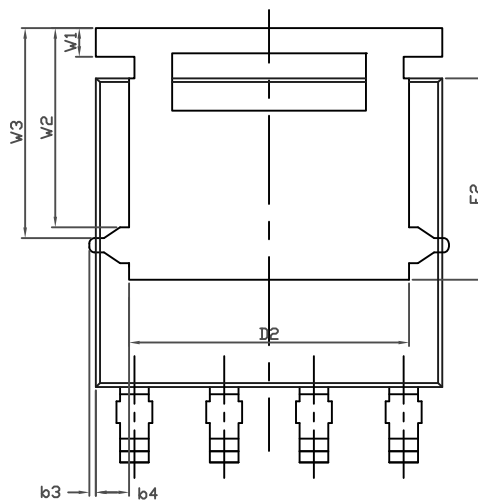
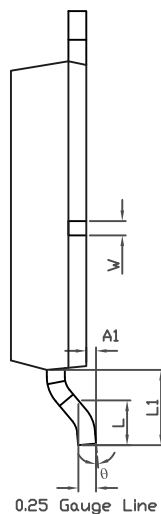
a. Old ordering code is obsolete and no longer valid for new orders



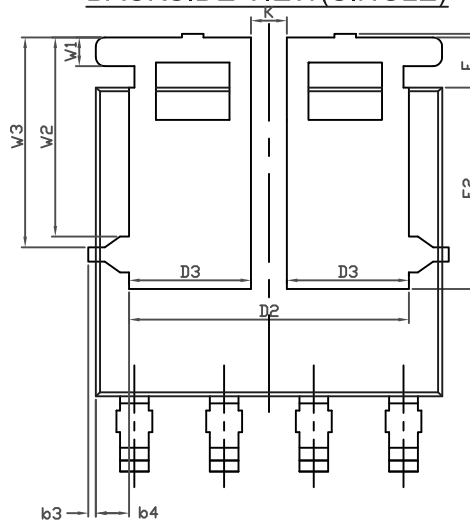
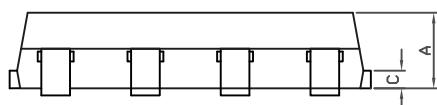
PowerPAK® SO-8L Case Outline for Non-AI Parts



TOPSIDE VIEW



BACKSIDE VIEW(SINGLE)



BACKSIDE VIEW(DUAL)



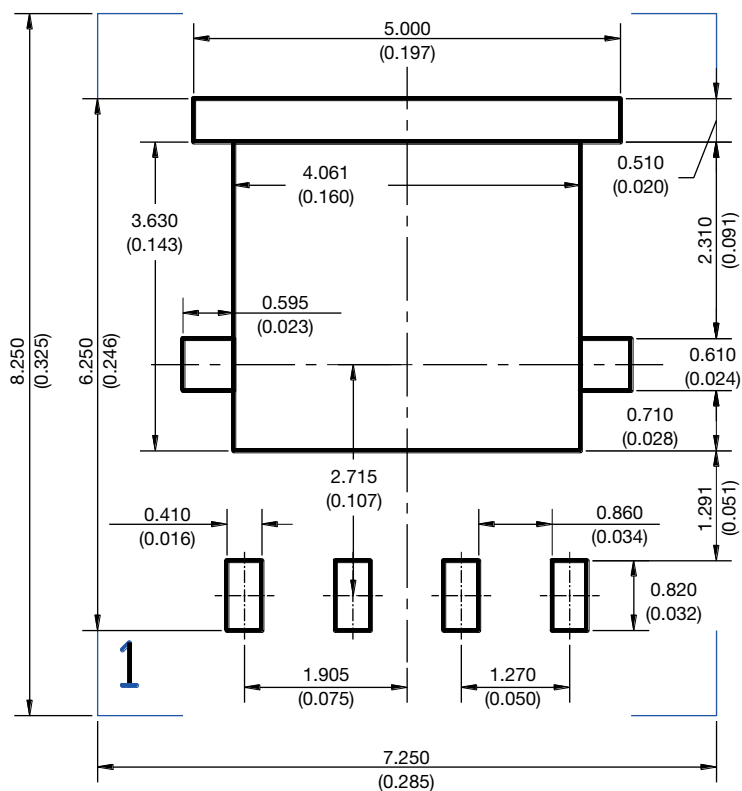
DIM.	MILLIMETERS			INCHES		
	MIN.	NOM.	MAX.	MIN.	NOM.	MAX.
A	1.00	1.07	1.14	0.039	0.042	0.045
A1	0.00	-	0.127	0.00	-	0.005
b	0.33	0.41	0.48	0.013	0.016	0.019
b1	0.44	0.51	0.58	0.017	0.020	0.023
b2	4.80	4.90	5.00	0.189	0.193	0.197
b3	0.094			0.004		
b4	0.47			0.019		
c	0.20	0.25	0.30	0.008	0.010	0.012
D	5.00	5.13	5.25	0.197	0.202	0.207
D1	4.80	4.90	5.00	0.189	0.193	0.197
D2	3.86	3.96	4.06	0.152	0.156	0.160
D3	1.63	1.73	1.83	0.064	0.068	0.072
e	1.27 BSC			0.050 BSC		
E	6.05	6.15	6.25	0.238	0.242	0.246
E1	4.27	4.37	4.47	0.168	0.172	0.176
E2	3.18	3.28	3.38	0.125	0.129	0.133
F	-	-	0.15	-	-	0.006
L	0.62	0.72	0.82	0.024	0.028	0.032
L1	0.92	1.07	1.22	0.036	0.042	0.048
K	0.51			0.020		
W	0.23			0.009		
W1	0.41			0.016		
W2	2.82			0.111		
W3	2.96			0.117		
q	0°	-	10°	0°	-	10°
ECN: C15-1122-Rev. C, 07-Sep-15 DWG: 5976						

Note

- Millimeters will govern



RECOMMENDED MINIMUM PAD FOR PowerPAK® SO-8L SINGLE



Recommended Minimum Pads
Dimensions in mm (inches)



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Vishay Intertechnology, Inc. hereby certifies that all its products that are identified as Halogen-Free follow Halogen-Free requirements as per JEDEC JS709A standards. Please note that some Vishay documentation may still make reference to the IEC 61249-2-21 definition. We confirm that all the products identified as being compliant to IEC 61249-2-21 conform to JEDEC JS709A standards.

Mouser Electronics

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Vishay:

[SQJ461EP-T1-GE3](#) [SQJ461EP-T1_GE3](#)